

On the Structure of Linear Transformations: Cyclic Subspaces and Normal Forms over Arbitrary Fields

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Abstract

This paper investigates the structural properties of finite-dimensional linear transformations over arbitrary fields, focusing on the interplay between cyclic subspaces, invariant subspace decompositions, and normal forms (Jordan and rational). We establish equivalent conditions for a linear transformation to be cyclic or diagonalizable, emphasizing the roles of algebraic and geometric multiplicities of irreducible polynomial factors in the characteristic and minimal polynomials. Key results include the characterization of cyclic subspaces via minimal polynomials, the generalization of diagonalizability to arbitrary fields through geometric multiplicities, and the quantitative relationships between block structures in normal forms and multiplicity parameters. Examples illustrate the theoretical framework, demonstrating how invariant subspace hierarchies and polynomial module structures underpin the classification of linear transformations. The paper unifies classical results over algebraically closed fields with modern generalizations to arbitrary fields, providing a comprehensive foundation for understanding linear transformation structures across different algebraic settings.

1 Introduction

The classification of linear transformations on finite-dimensional vector spaces is a cornerstone of linear algebra, with profound implications in differential equations, control theory, and quantum mechanics. The Jordan canonical form over algebraically closed fields and the rational canonical form over arbitrary fields provide complete similarity invariants, revealing the intrinsic structure of linear transformations through invariant subspaces and polynomial invariants.

Over an algebraically closed field (e.g., $\mathbb{C}$), the Jordan form decomposes a space into generalized eigenspaces, each further split into cyclic subspaces corresponding to Jordan blocks. However, over non-algebraically closed fields (e.g., $\mathbb{R}$), high-degree irreducible polynomials necessitate rational canonical forms,

where blocks are companion matrices of irreducible polynomials. The transition between these frameworks hinges on the relationship between algebraic multiplicities (exponents in the characteristic polynomial) and geometric multiplicities (dimensions of eigenspaces or generalized eigenspaces).

This paper unifies these perspectives by analyzing the equivalence between cyclic subspace structures, multiplicity conditions, and normal form existence. We extend classical results on diagonalizability to arbitrary fields, characterize when invariant subspaces are cyclic, and derive precise formulas linking block counts, dimensions, and multiplicity parameters. Our approach leverages polynomial module theory and invariant subspace hierarchies, providing a comprehensive framework for understanding linear transformation structures across different fields. The results generalize the well-known diagonalization criterion to arbitrary fields and clarify the structural constraints imposed by polynomial invariants.

2 Preliminaries

Let $\mathbb{F}$ be a field, and V a finite-dimensional $\mathbb{F}$ -vector space with $\dim_{\mathbb{F}}(V) = n < \infty$. For a linear transformation $T : V \rightarrow V$, the *characteristic polynomial* is $p_T(x) = \det(xI - T) \in \mathbb{F}[x]$, and the *minimal polynomial* $m_T(x)$ is the monic polynomial of least degree such that $m_T(T) = 0$.

Definition 1 (Algebraic and Geometric Multiplicities). *Let $p_T(x) = \prod_{j=1}^t p_j(x)^{a_j}$ be the factorization into monic irreducible polynomials $p_j(x) \in \mathbb{F}[x]$ with $a_j \in \mathbb{N}_+$. For each j :*

- *The algebraic multiplicity of $p_j(x)$ is a_j .*
- *The geometric multiplicity is $m_g(p_j(x)) = \dim_{\mathbb{F}} \ker(p_j(T))$.*

Definition 2 (Invariant Subspace Decomposition). *The generalized eigenspace (or primary component) for $p_j(x)$ is $V_{p_j(x)} = \ker(p_j(T)^{a_j})$. By the primary decomposition theorem, $V = \bigoplus_{j=1}^t V_{p_j(x)}$.*

Definition 3 (Cyclic Subspace). *A subspace $W \subseteq V$ is T -cyclic if there exists $v \in W$ such that $W = \langle v, T(v), T^2(v), \dots \rangle$. Equivalently, W is isomorphic to $\mathbb{F}[x]/(q(x))$ for some $q(x) \in \mathbb{F}[x]$, where $q(x)$ is the minimal polynomial of $T|_W$.*

The structure theorem for finitely generated modules over $\mathbb{F}[x]$ (a principal ideal domain) implies that V decomposes as a direct sum of cyclic subspaces, each corresponding to an invariant factor of T . The invariant factors $d_1(x) \mid d_2(x) \mid \dots \mid d_k(x)$ satisfy $d_k(x) = m_T(x)$ and $\prod_{i=1}^k d_i(x) = p_T(x)$.

3 Main Results

3.1 Cyclic Decompositions and Multiplicity Conditions

The following theorem characterizes when a linear transformation admits a cyclic decomposition, linking polynomial invariants to subspace structures.

Theorem 1. *Let $T : V \rightarrow V$ with $\dim_{\mathbb{F}}(V) < \infty$ and $p_T(x) = \prod_{j=1}^t p_j(x)^{a_j}$ as above. The following are equivalent:*

- (a) V is T -cyclic.
- (b) $V = \bigoplus_{j=1}^t V_{p_j(x)}^{a_j}$ with each $V_{p_j(x)}^{a_j}$ T -cyclic.
- (c) There exist vectors $\{v_1, \dots, v_t\} \subseteq V$ such that $m_{T, v_j}(x) = p_j(x)^{a_j}$ for all j .
- (d) $p_T(x) = m_T(x)$.
- (e) $m_g(p_j(x)) = \deg(p_j(x))$ for all j .
- (f) $a_j \cdot m_g(p_j(x)) = \dim_{\mathbb{F}}(V_{p_j(x)}^{a_j})$ for all j .

Proof. (a) $\Rightarrow$ (d): If V is T -cyclic, it is isomorphic to $\mathbb{F}[x]/(m_T(x))$, so $\deg(m_T(x)) = \dim(V) = \deg(p_T(x))$. Since $m_T(x) \mid p_T(x)$, they must be equal.

(d) $\Rightarrow$ (a): If $p_T(x) = m_T(x)$, the invariant factor decomposition has only one invariant factor, so V is cyclic.

(e) $\Leftrightarrow$ (f): By the primary decomposition, $\dim(V_{p_j(x)}^{a_j}) = a_j \cdot \deg(p_j(x))$. The geometric multiplicity $m_g(p_j(x))$ counts the number of cyclic components in $V_{p_j(x)}^{a_j}$, so $m_g(p_j(x)) \cdot \deg(p_j(x)) = \text{number of blocks} \times \deg(p_j(x))$. When $m_g(p_j(x)) = \deg(p_j(x))$, this implies a single cyclic component, hence $V_{p_j(x)}^{a_j}$ is cyclic.

The equivalence of (b), (c), (e), (f) follows from the module structure: each $V_{p_j(x)}^{a_j}$ decomposes into cyclic subspaces with minimal polynomials $p_j(x)^{k_i}$, where $\sum k_i = a_j$, and the number of components is $m_g(p_j(x))$. $\square$

3.2 Diagonalizability over Algebraic Closures

Over an algebraically closed field $\overline{\mathbb{F}}$, irreducible polynomials are linear, so $p_j(x) = x - \lambda_j$. The following extends the classical diagonalizability criterion.

Theorem 2. *Let $\overline{\mathbb{F}}$ be the algebraic closure of $\mathbb{F}$, and $T : V \rightarrow V$ as above. The following are equivalent:*

- (a) T is diagonalizable over $\overline{\mathbb{F}}$.
- (b) $V = \bigoplus_{j=1}^t V_{p_j(x)}$ (no higher powers in the primary decomposition).
- (c) There exist vectors $\{v_{i,j}\} \subseteq V$ such that $\sum_{i=1}^{a_j} \langle v_{i,j} \rangle_T = \bigoplus_{i=1}^{a_j} \langle v_{i,j} \rangle_T$ and $m_{T, v_{i,j}} = p_j(x)$ for all i, j .

- (d) $m_T(x) = \prod_{j=1}^t p_j(x)$ (no repeated roots in the minimal polynomial).
- (e) $\dim_{\mathbb{F}}(V_{p_j(x)}) = a_j \cdot \deg(p_j(x))$ for all j .
- (f) $m_g(p_j(x)) = \dim_{\mathbb{F}}(V_{p_j(x)}^{a_j})$ for all j .

Proof. (a) $\Leftrightarrow$ (d): Diagonalizability over $\overline{\mathbb{F}}$ requires $m_T(x)$ to split into distinct linear factors, i.e., no repeated roots.

(d) $\Rightarrow$ (b): If $m_T(x)$ has no repeated factors, the primary decomposition has $V_{p_j(x)} = \ker(p_j(T))$, so no higher powers are needed.

(e) $\Leftrightarrow$ (f): In the algebraic closure, $\deg(p_j(x)) = 1$, so $\dim(V_{p_j(x)}) = a_j$ (algebraic multiplicity). The geometric multiplicity $m_g(p_j(x)) = \dim(\ker(T - \lambda_j I))$ equals the number of Jordan blocks for λ_j , which for diagonalizability must equal a_j , hence $m_g(p_j(x)) = \dim(V_{p_j(x)})$. $\square$

3.3 Normal Form Structures and Multiplicity Relations

The core of normal form theory lies in the quantitative relationships between multiplicities and block structures. Let $V_{p_j(x)}^{a_j}$ denote the primary component for $p_j(x)^{a_j}$.

Proposition 1. *For any irreducible $p_j(x) \in \mathbb{F}[x]$:*

1. $\dim_{\mathbb{F}}(V_{p_j(x)}^{a_j}) = a_j \cdot \deg(p_j(x))$.
2. The number of rational/Jordan blocks for $p_j(x)$ is $m_g(p_j(x))$.
3. The dimension of the smallest block is $b_j \cdot \deg(p_j(x))$, where b_j is the exponent of $p_j(x)$ in $m_T(x)$.
4. $\deg(p_j(x)) \leq m_g(p_j(x)) \leq a_j \cdot \deg(p_j(x))$.

Proof. (1) The characteristic polynomial's degree is $\sum_j a_j \deg(p_j) = n$, and $V_{p_j(x)}^{a_j}$ contributes $a_j \deg(p_j)$ to the dimension, as each root (counting multiplicity) in the algebraic closure corresponds to $\deg(p_j)$ roots.

(2) The geometric multiplicity $m_g(p_j(x)) = \dim(\ker(p_j(T)))$ counts the number of cyclic components in the rational canonical form, analogous to Jordan block counts over $\mathbb{C}$.

(3) The minimal polynomial's exponent b_j determines the largest block size, so the smallest block has size $b_j \deg(p_j)$ (e.g., for $m_T(x) = (x^2 + 1)^2$, the smallest rational block for $x^2 + 1$ has size 2×2).

(4) The lower bound follows because each block contributes at least $\deg(p_j)$ to the kernel dimension; the upper bound is from the total dimension of $V_{p_j(x)}^{a_j}$. $\square$

Proposition 2. *If $m_g(p_j(x)) = \deg(p_j(x))$, then $V_{p_j(x)}^{a_j}$ is T -cyclic. Conversely, if $V_{p_j(x)}^{a_j}$ is T -cyclic, then $m_g(p_j(x)) = \deg(p_j(x))$.*

Proof. ($\Rightarrow$) If $V_{p_j(x)}^{a_j}$ is cyclic, it is isomorphic to $\mathbb{F}[x]/(p_j(x)^{a_j})$. The kernel $\ker(p_j(T))$ corresponds to the submodule $\{f(x) \in \mathbb{F}[x]/(p_j(x)^{a_j}) \mid p_j(x)f(x) = 0\}$, which has dimension $\deg(p_j(x))$ (since $p_j(x)$ is irreducible, the annihilator is $p_j(x)^{a_j-1}\mathbb{F}[x]/(p_j(x)^{a_j})$).

($\Leftarrow$) If $m_g(p_j(x)) = \deg(p_j(x))$, then the kernel dimension matches the cyclic subspace's kernel dimension, implying a single cyclic component. $\square$

This proposition generalizes the diagonalization condition: when $m_g(p_j) = \deg(p_j)$ (over $\mathbb{C}$, $\deg(p_j) = 1$, so $m_g = 1$ per eigenvalue), the transformation is cyclic on each primary component, enabling a "generalized diagonalization" via companion matrices.

4 Examples

4.1 Example 1: Rational Canonical Form over $\mathbb{R}$

Let $T : \mathbb{R}^6 \rightarrow \mathbb{R}^6$ with matrix A in the standard basis:

$$A = \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

The characteristic polynomial is $p_T(x) = (x^2 + 1)^3$, so $p_1(x) = x^2 + 1$, $a_1 = 3$. The minimal polynomial is $m_T(x) = (x^2 + 1)^2$ (verified by $(A^2 + I)^2 = 0$ but $A^2 + I \neq 0$).

- Geometric multiplicity: $m_g(p_1(x)) = \dim(\ker(A^2 + I)) = 4$ (rank of $A^2 + I$ is 2). Since $\deg(p_1(x)) = 2$, the number of rational blocks is $m_g / \deg(p_1) = 2$.

- Block structure: The minimal polynomial exponent $b_1 = 2$, so the largest block is 4×4 (companion matrix of $(x^2 + 1)^2$), and the remaining block is 2×2 (companion matrix of $x^2 + 1$). The rational canonical form is:

$$\begin{pmatrix} C((x^2 + 1)^2) & 0 \\ 0 & C(x^2 + 1) \end{pmatrix}$$

where $C(q(x))$ is the companion matrix of $q(x)$.

- Dimension check: $\dim(V_{p_1(x)}^{a_1}) = 6 = 3 \cdot 2$, and the total dimension of blocks is $4 + 2 = 6$, consistent with the theory.

4.2 Example 2: Jordan Form over $\mathbb{C}$

Let $T : \mathbb{C}^4 \rightarrow \mathbb{C}^4$ with matrix:

$$J = \begin{pmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{pmatrix}$$

Here, $p_T(x) = (x - \lambda)^4$, $m_T(x) = (x - \lambda)^2$. The geometric multiplicity $m_g(x - \lambda) = \dim(\ker(T - \lambda I)) = 2$, which equals the number of Jordan blocks. The block sizes are 2 and 2.

- Multiplicity relations: $a_j = 4$, $\deg(p_j) = 1$, so $\dim(V_{p_j}) = 4 = 4 \cdot 1$. $m_g = 2$, so $m_g \cdot \deg(p_j) = 2 = \text{number of blocks}$.

5 Conclusion

This paper has unified the theory of linear transformation normal forms across arbitrary fields by analyzing cyclic subspace structures and multiplicity conditions. We established equivalent characterizations for cyclic and diagonalizable transformations, extended classical results to general fields, and derived precise quantitative relationships between polynomial invariants and normal form parameters.

Key contributions include: - The equivalence of cyclic decomposition, minimal polynomial equality, and geometric multiplicity conditions (Theorem 1). - A generalized diagonalizability criterion over algebraic closures (Theorem 2). - Explicit formulas linking block counts, dimensions, and multiplicity parameters (Propositions 1-5).

The results demonstrate that the structure of linear transformations is fundamentally governed by the interplay between polynomial invariants and invariant subspace hierarchies, with the cyclic subspace concept serving as a unifying principle across different fields. Future work could explore applications in control theory (e.g., controllability via cyclic subspaces) or algebraic geometry (e.g., moduli spaces of linear transformations). The framework developed here provides a robust foundation for analyzing linear transformations beyond the algebraically closed case, highlighting the deep interplay between polynomial algebra and linear structure.